

A.I., MACHINE LEARNING, AND ETHICAL DECISION-MAKING IN SUSTAINABLE INFRASTRUCTURE: A HURRICANE SANDY MODELING PERSPECTIVE

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Abstract:

This study examines how modeling frameworks and variable selection influence disaster prioritization using the EM-DAT global disaster database. A Linear Regression model and Random Forest model were compared to evaluate how conventional impact indicators such as deaths, affected population, and economic damage, shape a composite severity score.

An ethical stress-test was conducted by removing dominant indicators to observe shifts in model behavior. Results show that models strongly prioritize measurable impacts when available but shift toward contextual variables such as duration and rare high-consequence events when those indicators are removed.

Hurricane Sandy (2012) is used as a regional case study demonstrating that disaster ranking varies depending on model structure and classification. These findings highlight that AI-based prioritization reflects structured analytical assumptions rather than objective representations of disaster severity.

Introduction:

- Disaster datasets rely on predefined classification systems (e.g., type, subtype, region) to organize complex real-world events. While useful for standardization, these classifications are simplified and may not fully capture real-world dynamics.
- As AI and machine learning models are increasingly used in disaster assessment, resource allocation, and infrastructure planning, these classification structures become embedded in analytical systems.

This raises a key question:
Do models reflect reality, or the way we structure reality?

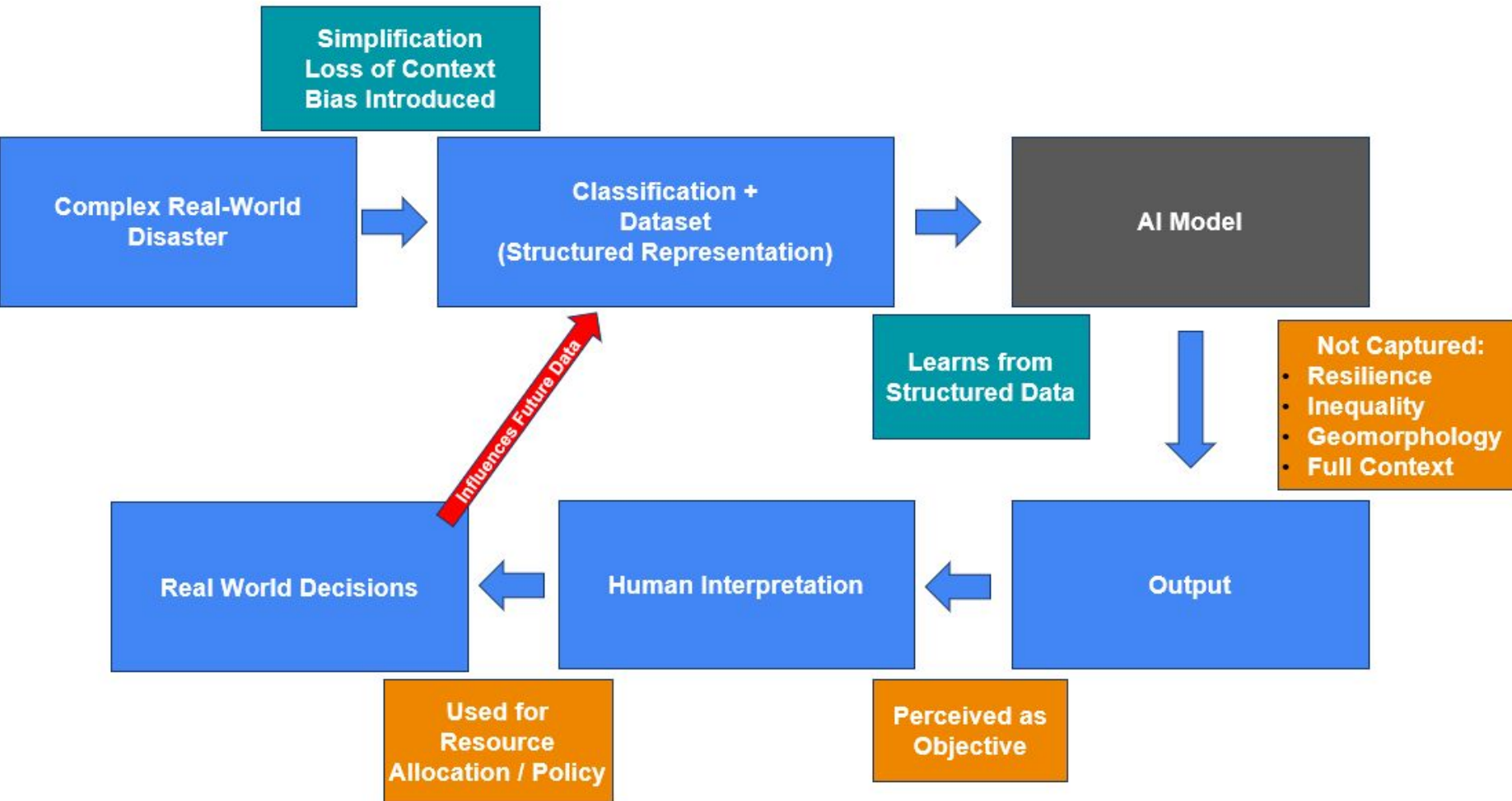


Figure 1: Conceptual framework illustrating how complex real-world disasters are translated into structured classifications and datasets that inform AI models. This transformation introduces simplification and loss of context, embedding bias into model outputs. These outputs can influence human interpretation, resource allocation, and future data representation.

Methods:

- Data Source:**
 - EM-DAT global disaster database (10,640 records)
- Processing:**
 - Log-transformed deaths, affected population, and economic damage
 - Composite severity score (standardized)
 - Derived variables: duration, data completeness, rare high-consequence events
- Models:**
 - Linear Regression
 - Random Forest
- Ethical Stress-Test**
 - Full model (includes deaths, affected, damage)
 - Reduced model (removes deaths, affected, damage)
 - Used to evaluate how model priorities shift

Results:

$$Ensemble\ Score = \alpha(AI\ Priority\ Score) + (1 - \alpha)(Human\ Ethical\ Score)$$

- Full Models:
- Prioritize measurable impacts:
 - affected population
 - deaths
 - economic damage

- Reduced Models:
- Shift toward contextual variables:
 - Duration
 - Rare high-consequence events
 - Data completeness
 - Regional context

Key Finding:
When dominant variables are removed, models re-prioritize proxy variables.

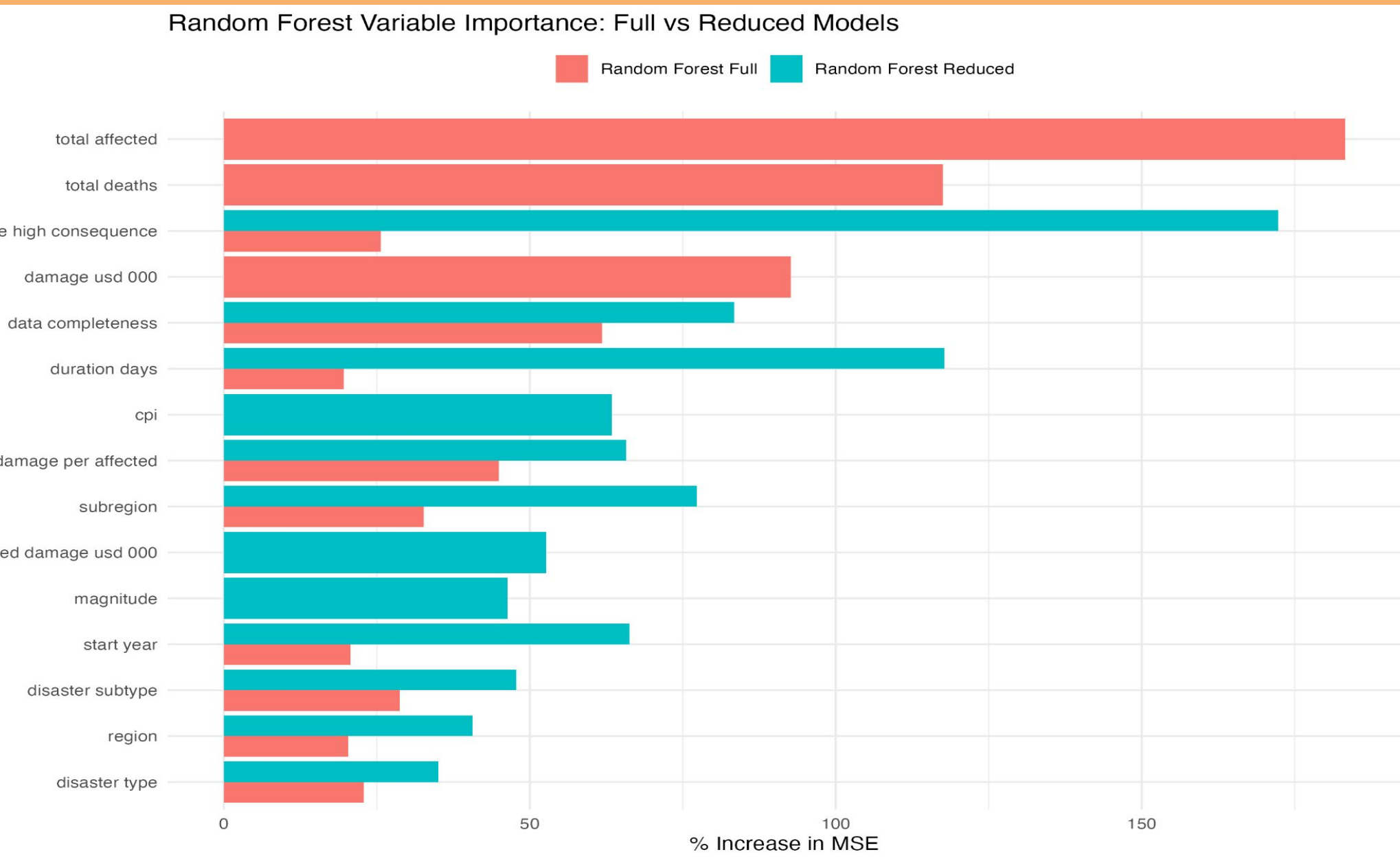


Figure 2: Random Forest variable importance comparison between full and reduced models. When dominant impact variables (e.g., deaths, affected population, damage) are removed, model importance shifts toward contextual indicators. This demonstrates that model-defined “severity” is contingent on available inputs.

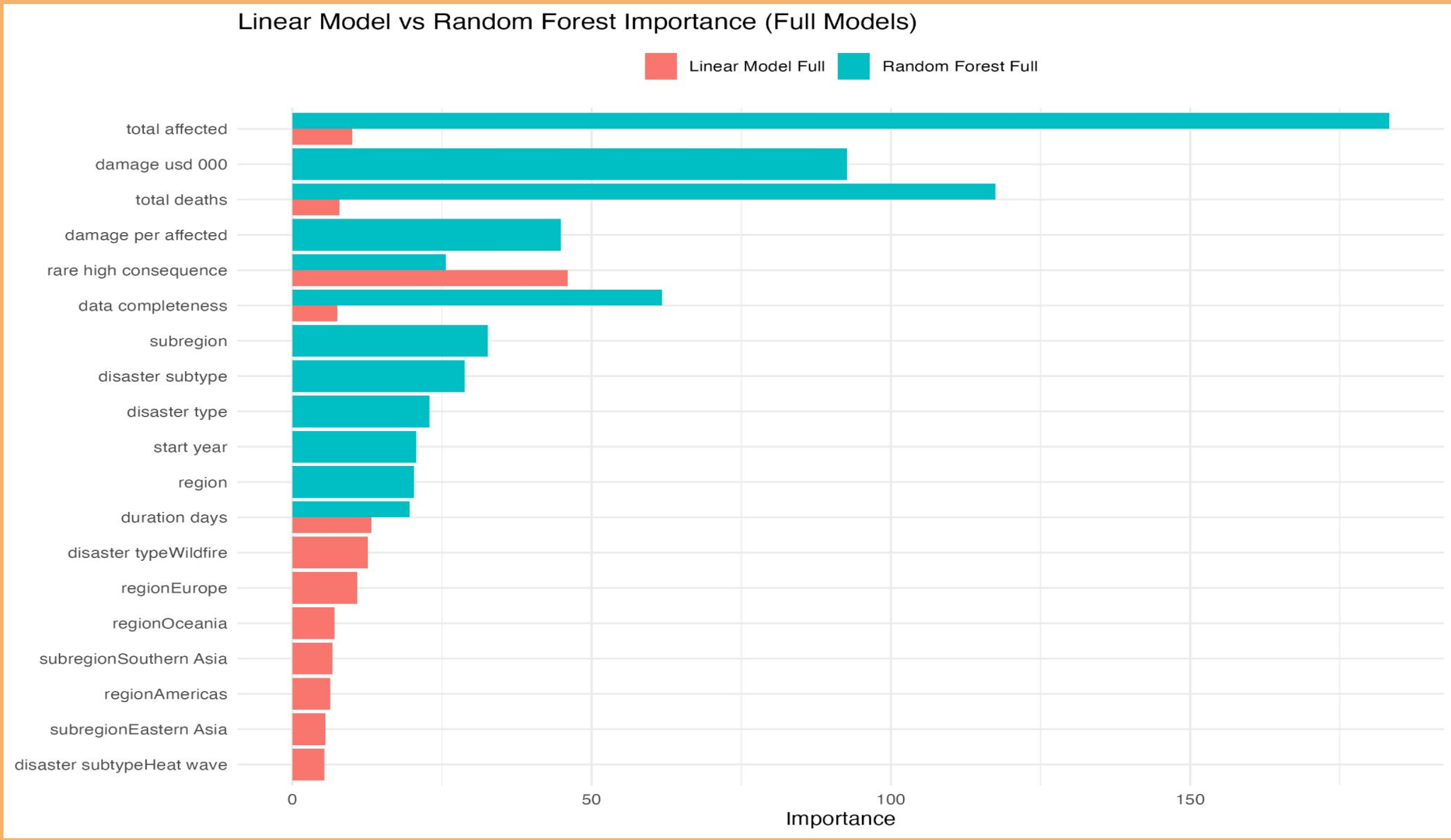


Figure 3: Comparison of variable importance across Linear Regression and Random Forest models using the same dataset. Differences in variable prioritization demonstrate that model structure influences how “importance” is defined, indicating that there is no single objective interpretation of disaster severity within algorithmic frameworks.

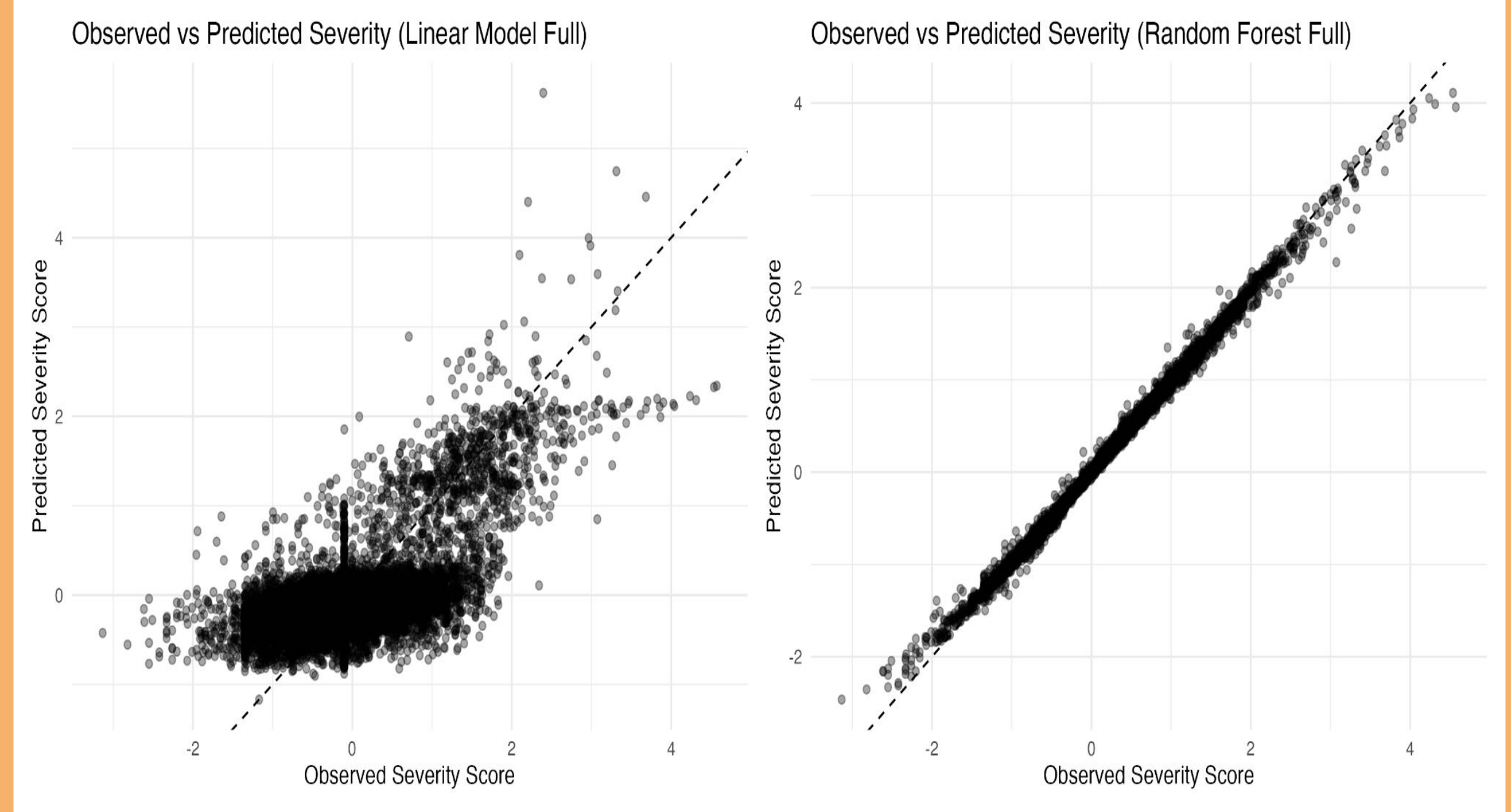


Figure 4: Observed versus predicted severity scores for Linear Regression and Random Forest models. While the Random Forest model demonstrates strong predictive accuracy, this performance reflects alignment with structured inputs rather than representation of real-world disaster complexity.

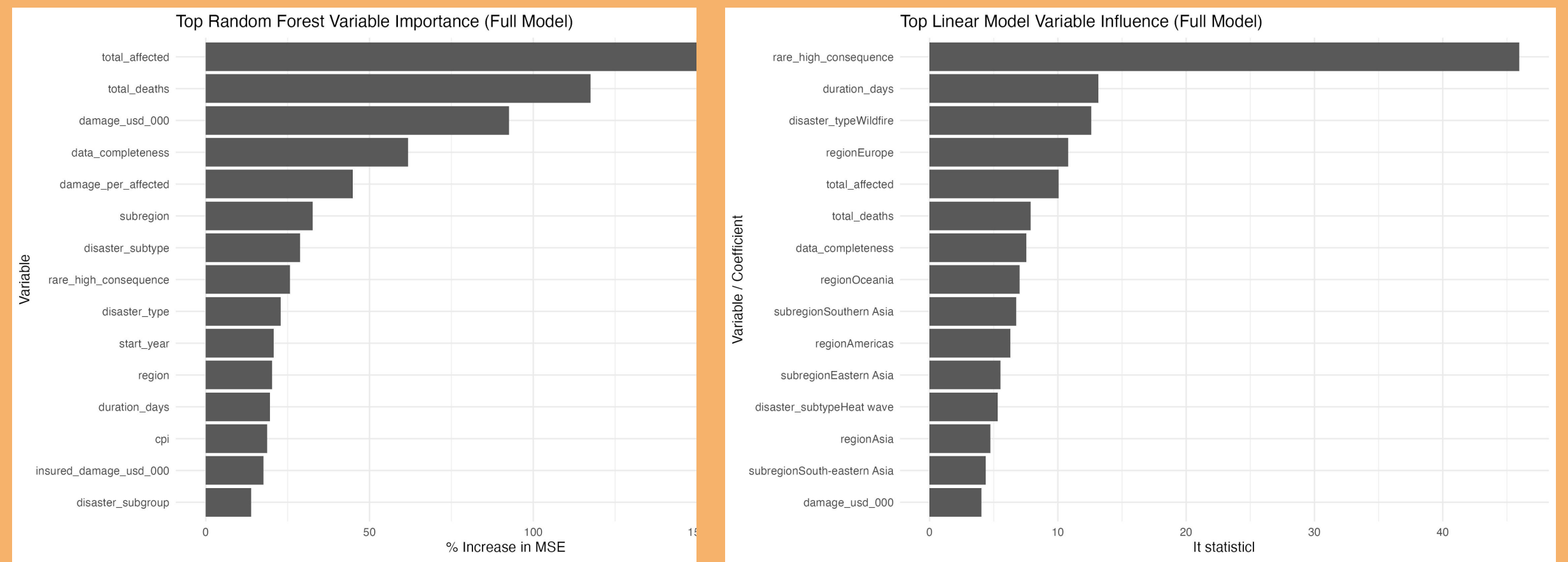


Figure 5: Feature importance from the linear model shows that rare, high-consequence events have the strongest influence on predictions, followed by duration and disaster type, highlighting how the model is heavily shaped by how the data is categorized and structured.

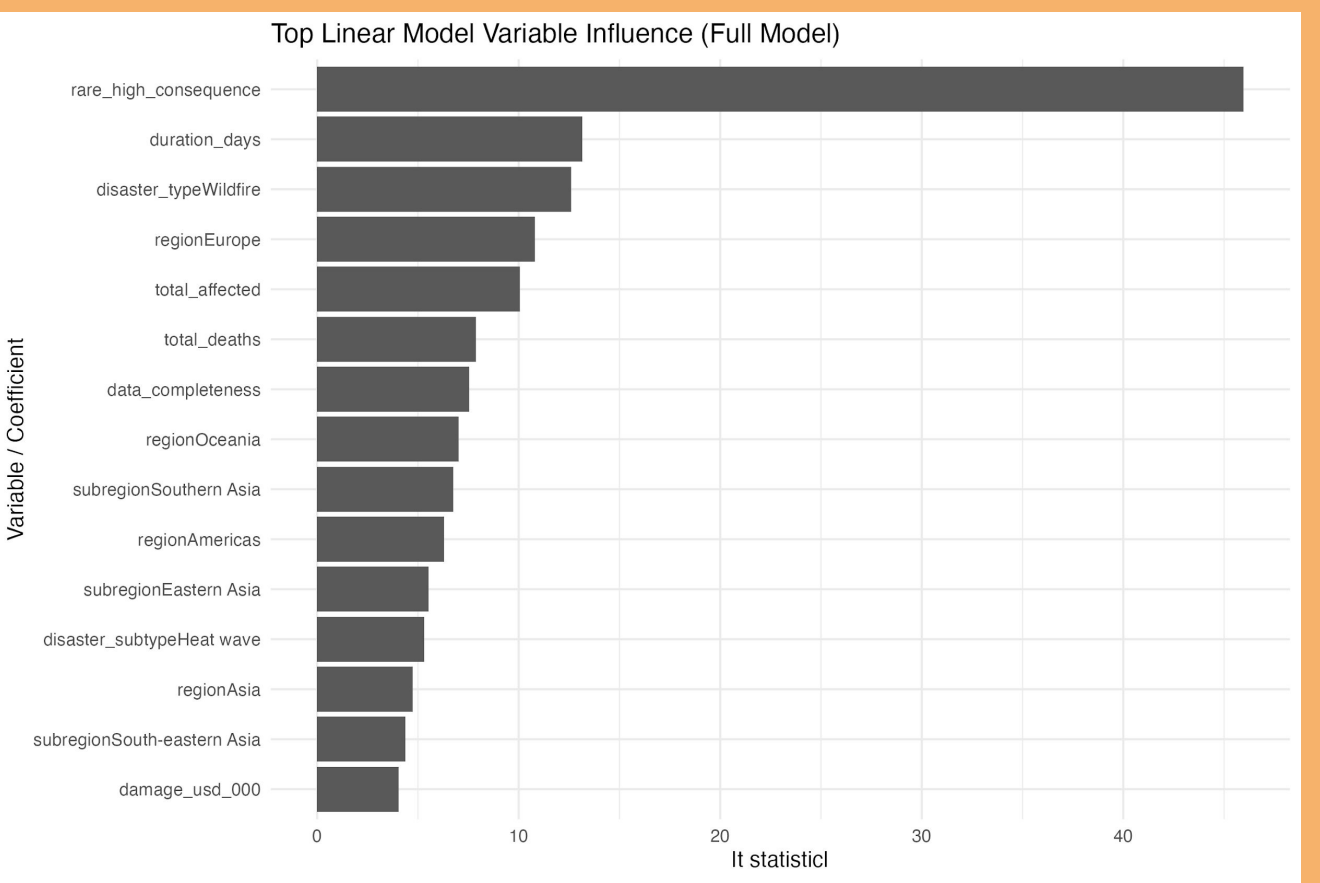


Figure 6: Random Forest feature importance shows that total affected has the strongest impact on predictions, followed by deaths and economic damage, indicating the model prioritizes large-scale measurable impacts over categorical labels.

Discussion/Conclusion:

AI models do not measure severity but rather operationalize it.

Severity is constructed through:

- Data availability
- Classification systems
- Model design

As a result:

- Measurable impacts are prioritized
- Contextual and structural factors may be underrepresented

AI-informed decisions may unintentionally reinforce structural bias, influencing resource allocation and policy decisions. The risk is not that models are incorrect, it's that they are incomplete.

Credit Authorship Contribution Statement:

Izaguirre Cortez, B.: RStudio coding, conceptualization, methodology, model development, visualization, and paper composition. Rezki, A.: review, editing, and references. Najarro Lima, M.: review and editing. Marsellos, A. E.: supervision, conceptual guidance, contribution to the Hurricane Sandy case study and Discussion, regional context, references, review and editing.

Acknowledgements:

Special thanks to Dr. Antonios E. Marsellos and the Department of Geology, Environment and Sustainability for their support and guidance in this research project.

References:

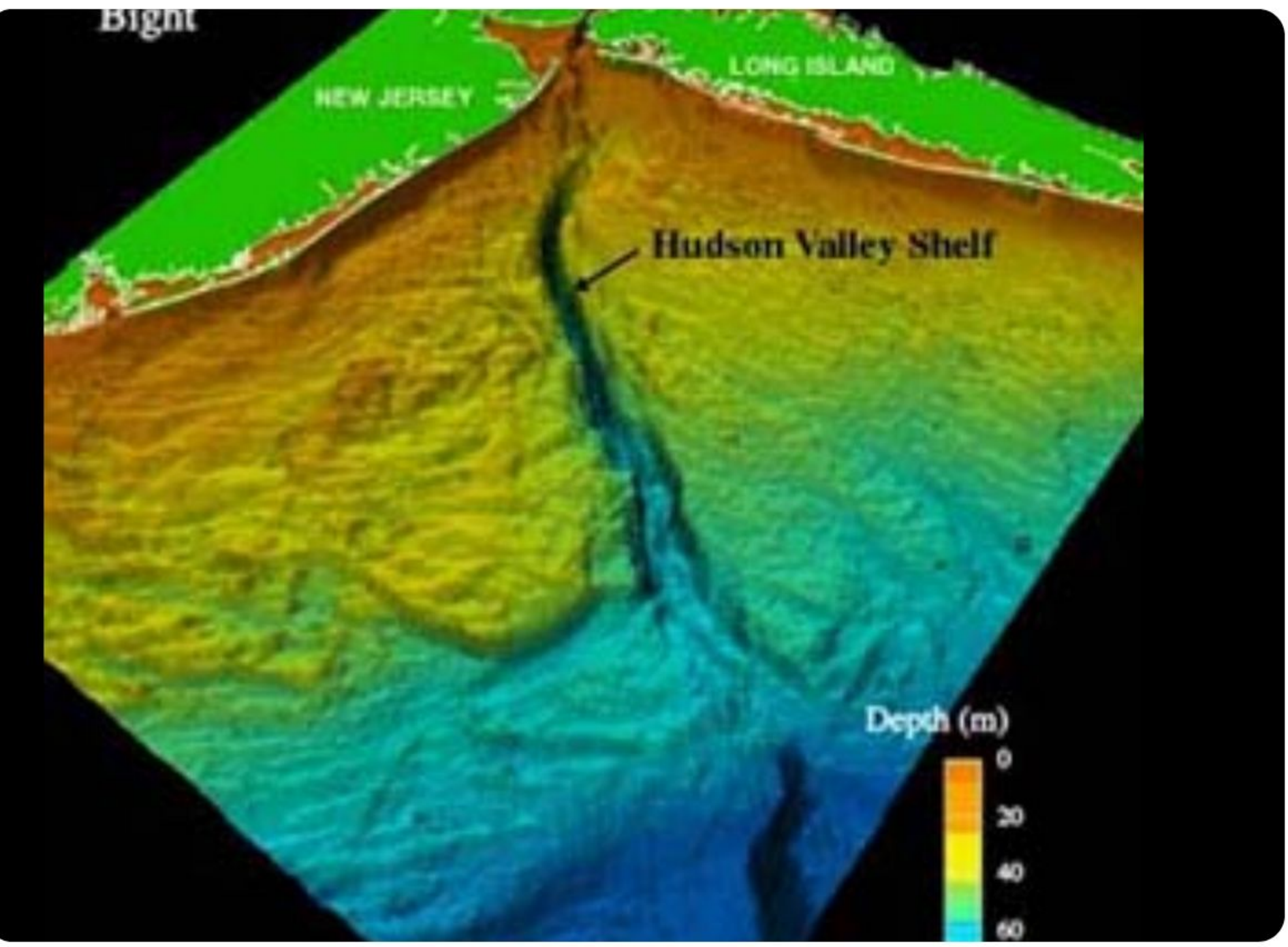


Figure 7: New York Bight Geomorphology, The concave coastline and shallow shelf amplified the surge.

Source: https://en.wikipedia.org/wiki/New_York_Bight